Sales Performance Analysis - 2025

Python Data Analysis Project | January - December 2025

Executive Summary

This report analyzes 1,000 sales orders from 2025 to evaluate revenue, profitability, product performance, sales channels, customer segments, and geographic trends.

Key Performance Indicators

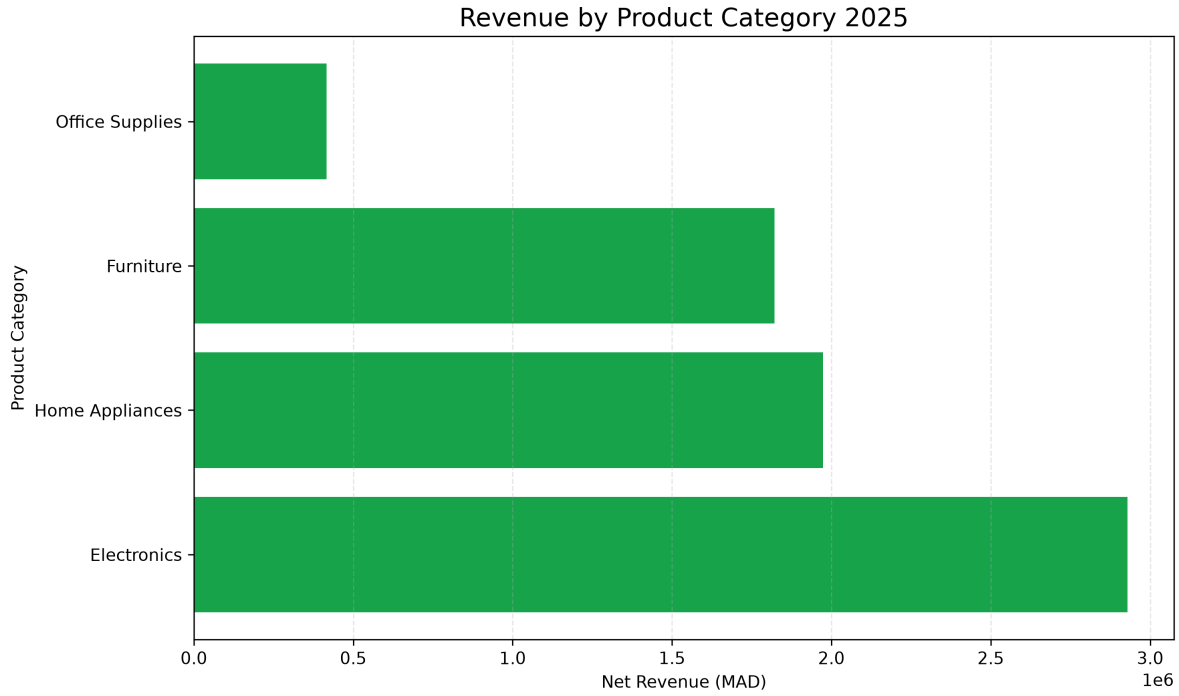
Metric	Value
Total Revenue	7,139,581.00 MAD
Total Profit	1,900,950.98 MAD
Profit Margin	26.63%
Total Orders	1,000

Data Preparation

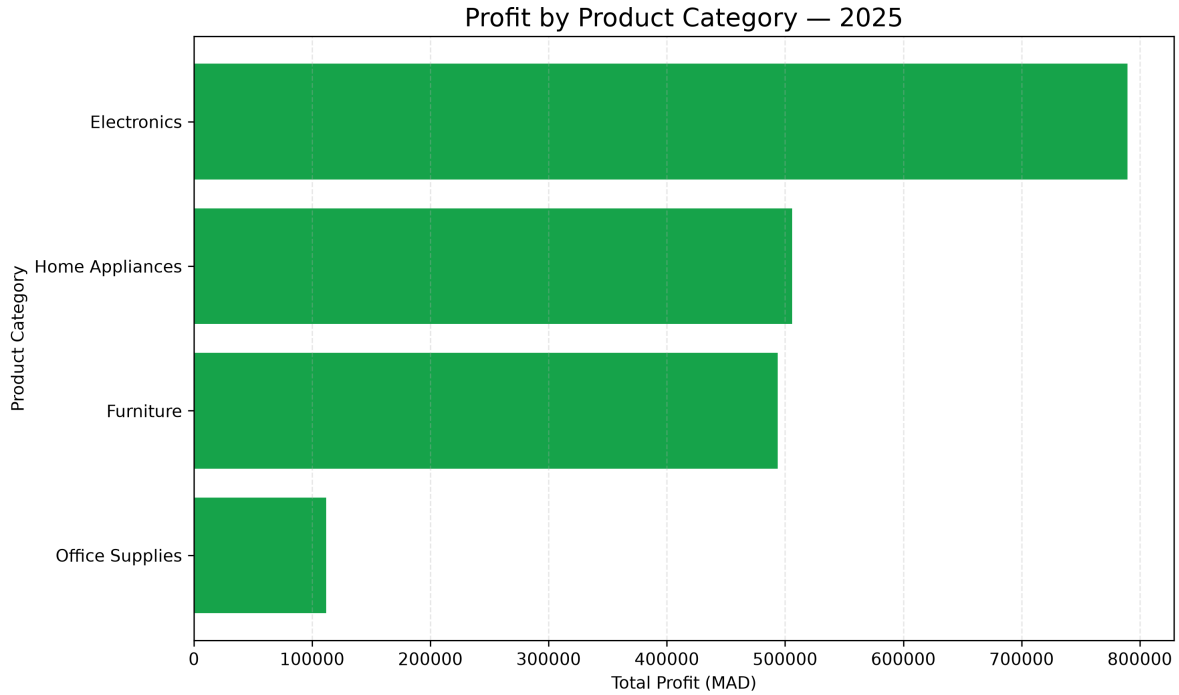
The original dataset contained 1,005 rows. Five duplicate orders were removed, resulting in 1,000 unique orders. Missing Customer_ID and City values were labeled as Unknown. Data quality checks covered quantities, revenue, profit calculations, discounts, dates, and duplicate order IDs.

Sales Performance Visualizations

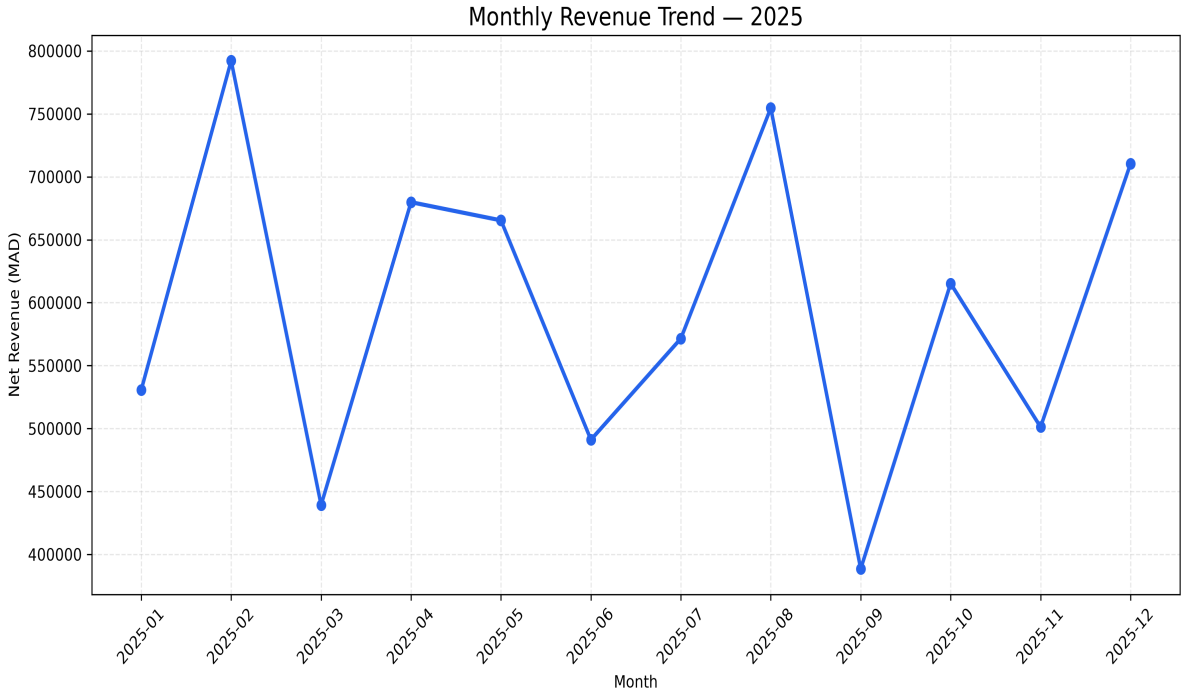
Revenue by Product Category



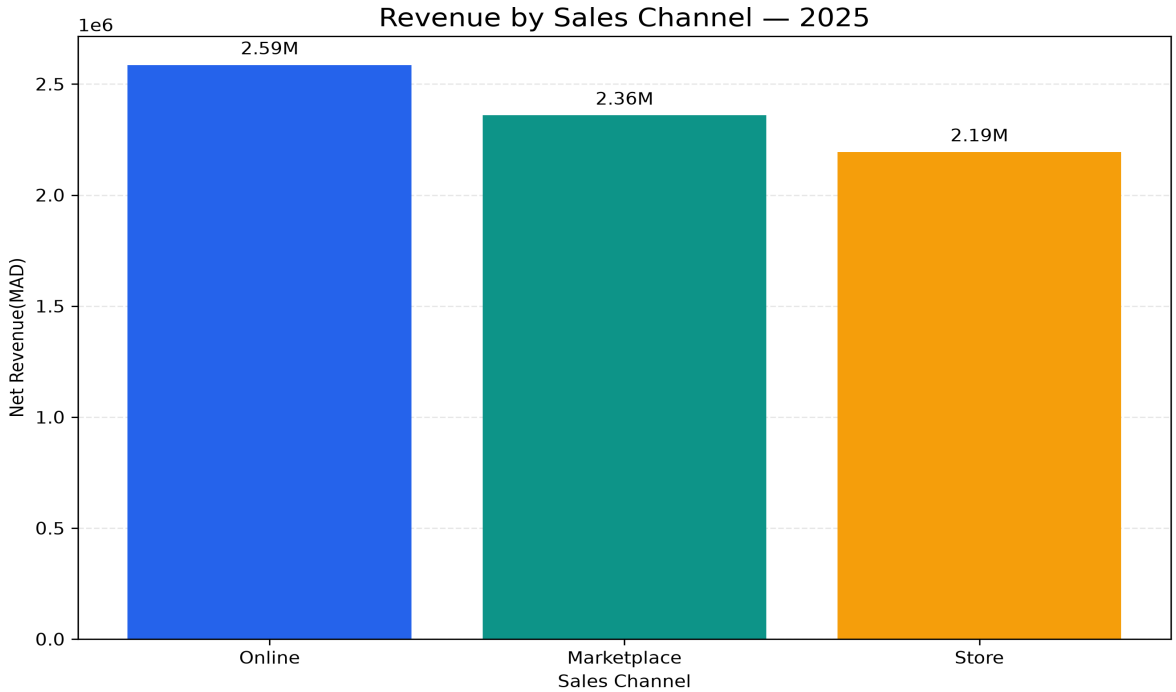
Profit by Product Category



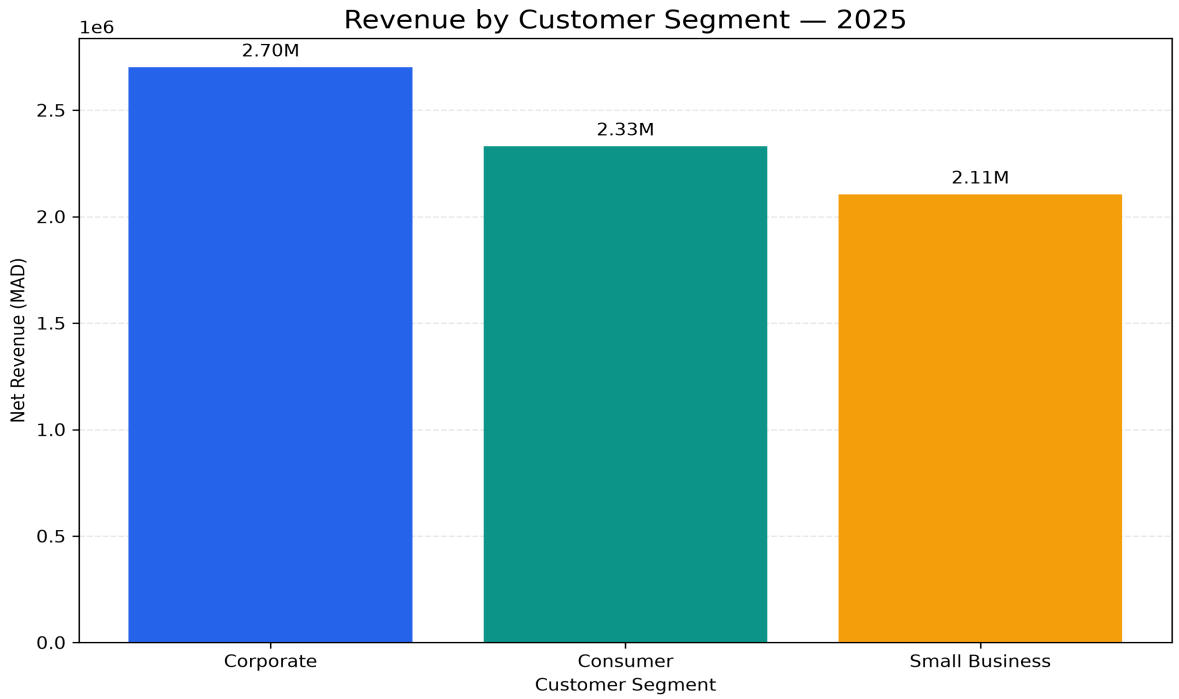
Monthly Revenue Trend



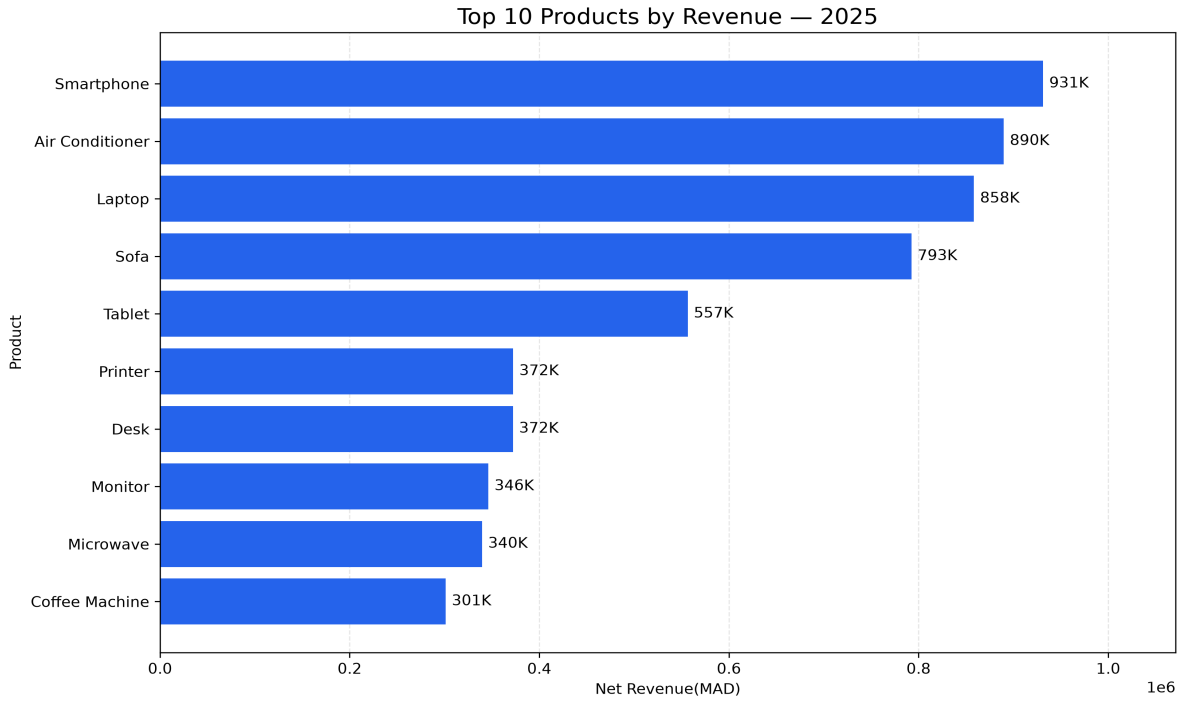
Revenue by Sales Channel



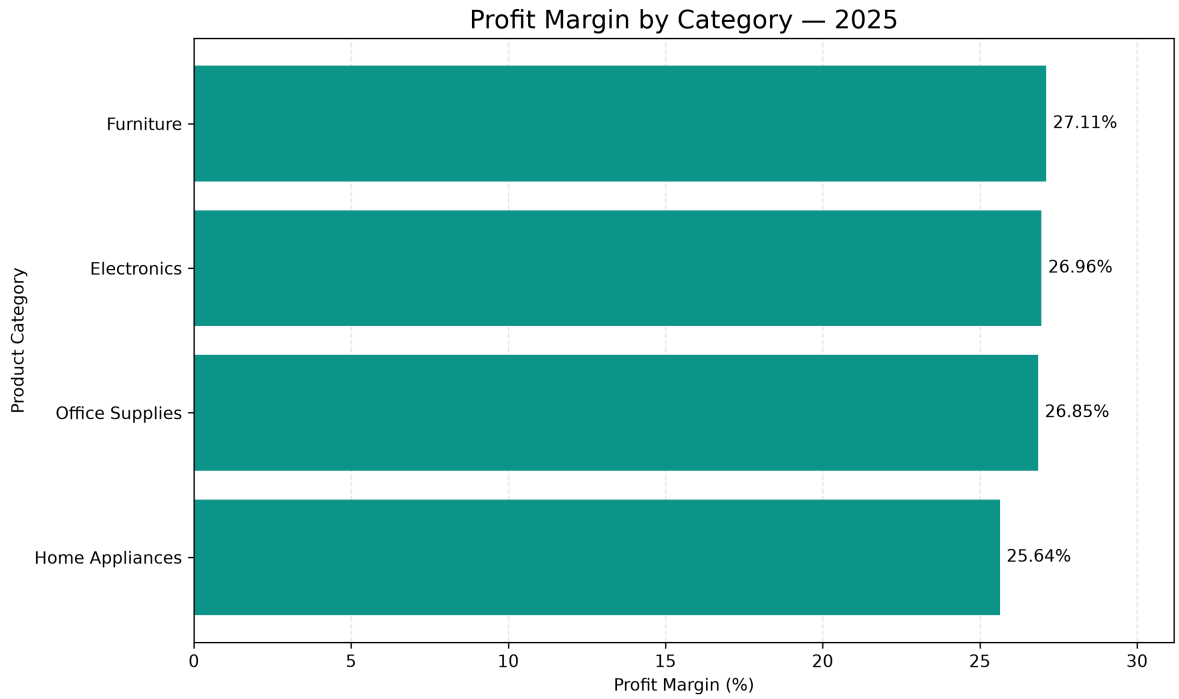
Revenue by Customer Segment



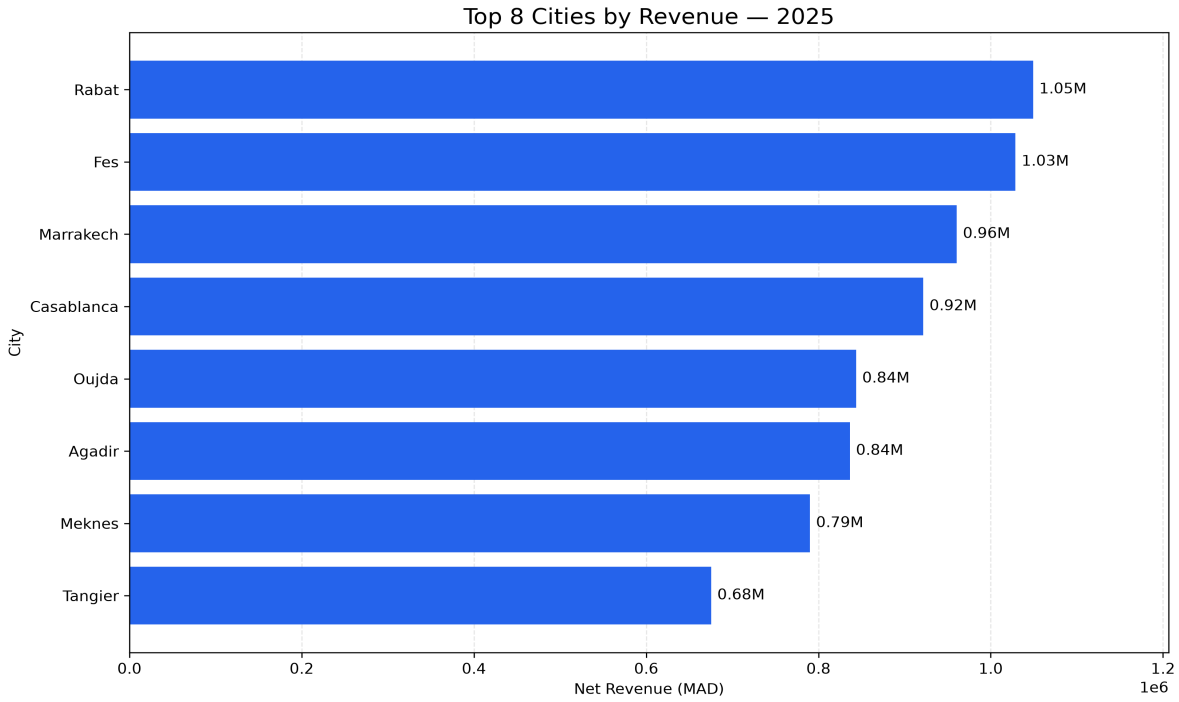
Top 10 Products by Revenue



Profit Margin by Category



Top 8 Cities by Revenue



Key Business Insights

Category Performance

Electronics generated the highest revenue (2.93M MAD) and profit (789K MAD). Furniture achieved the highest profit margin (27.11%).

Product Performance

Smartphone was the leading product by revenue (931K MAD), followed by Air Conditioner (890K MAD) and Laptop (858K MAD).

Geographic Performance

Rabat generated the highest city revenue (1.05M MAD), followed by Fes and Marrakech.

Sales Channel Performance

Online generated the highest revenue (2.59M MAD), while Marketplace achieved the highest profit margin (27.35%).

Customer Segment Performance

Corporate generated the highest revenue (2.70M MAD) and profit margin (27.90%).

Monthly Performance

February recorded the highest monthly revenue (792K MAD), while September recorded the lowest (388K MAD).

Business Recommendations

1. Review inventory and demand for top-selling products.

2. Investigate cost and discount differences across categories.

3. Analyze Marketplace profitability before expanding investment.

4. Evaluate targeted opportunities for Corporate customers.

5. Investigate the September revenue decline and October recovery.

Analysis Limitations

- The dataset covers only 2025.

- Five orders have unknown city information.

- The findings do not establish causation.

- Business recommendations require further validation.